SOUND OF URBAN ENVIRONMENT - TODAY AND FUTURE

Sze Wing Chan
KTH Royal Institute of Technology
Stockholm, Sweden
swch@kth.se

Wei-Jun Lin
KTH Royal Institute of Technology
Stockholm, Sweden
wjlin@kth.se

ABSTRACT

Urban sounds are rich yet often overlooked aspects of city life, continuously shaped by technological, social, and environmental change. This project investigates how immersive media—particularly augmented reality (AR) and spatial audio– can be used to reflect on and re-imagine the of soundscapes of urban environments. With the use of design fiction methodologies, we created soundscapes that illustrates the present and speculative future of the KTH open-air square in Stockholm through a combination of real audio recordings, synthesized sounds, and layered sound design. These soundscapes are placed in an interactive AR installation where audiences can navigate a physical space filled with virtual AR windows representing each corner of the open-air square. These windows allow users to hear the current and future soundscapes of how urban environments may evolve by 2040. The experience invites reflection on urban futures, highlighting how sound can serve as a powerful medium for storytelling and participatory imagination for the future.

1. INTRODUCTION

Sound is a vital element of how we experience and understand urban environments. While visual cues dominate our perception of space, soundscapes carry equally rich information about activity, culture, atmosphere, and time. From the hum of traffic to the chatter of pedestrians or the distant chirping of birds, our surrounding sounds shape our sense of place and community. However, urban soundscapes are not static—they evolve with technological advancement, shifts in social patterns, environmental changes, and more. Despite the richness of our sonic environments, we rarely pause to reflect on the soundscapes we inhabit in or how they might change in the future as urban sound tends to be treated as background noise.

At the same time, immersive media technologies—such as augmented reality (AR) and spatial audio—are provide new opportunities on how we engage with digital content in physical space. These tools offer new possibilities for embodied interaction, making it possible to situate digital sound in physical spaces and invite users to explore sonic environments in intuitive ands patial ways. Moreover, design fiction is increasingly used to explore future possibilities, with a rise in it its application of exploring the future of urban space through sound [1, 2].

This project hopes to bring attention to the evolving nature of urban soundscapes and to use sound as a medium for reflection, speculation, and engagement. We were interested in how sound can embody change—not only in what we hear but in how we think about our surroundings. Using design fiction methods and AR and spatial audio tools, we aimed to create an interactive experience that invites audiences to listen actively, reflect on the present, and imagine possible futures.

In order to do so, we created an interactive AR installation that allows audiences to explore an indoor room with virtual AR objects, each representing a specific corner of the KTH open-air square. Through the experience, audiences can listen and compare two types of soundscapes of the open-air square—one capturing its present acoustic character, the other imagining how it might sound in a near possible future in 2040. These experiences are designed to prompt reflection on how urban life will change in the future through sound and to explore how the combination of immersive technologies and spatial audio can mediate our perception of time, space, and change.

2. BACKGROUND

This section provides an overview of the theories and methodologies used in this project. It begins by outlining key concepts and frameworks in design fiction, followed by an examination of methodologies used to construct narratives of speculative futures. Afterward, it explores the role of sound in enhancing immersive experiences, focusing on sound design principles, categorization of urban sounds, and sound localization

2.1 Design Fiction

Design fiction is a creative approach that uses design to explore possible futures. This term was coined by Sterling, which he defined as ”the deliberate use of diegetic prototypes to suspend disbelief about change” [3]. It is later developed by Bleecker [4], where he states design fiction combines elements of design, science fact, and science fiction to create storytelling artifacts that materialize the imagination of the future while allowing space for discussions and speculations about those imaginations. In other words, design fiction, instead of predicting how things will be, it asks how things could be to open up different discussions and interpretations.

Based on Sterling and Bleecker’s idea, Lindley and Coulton [5] presented three elements of design fiction–(1) something that creates a story world, (2) something being prototypes within the story world created in (1), and (3) creates

a discursive space for discussion. They also stated that design fiction's nature is flexible, which leads to ambiguity. In which this ambiguity provides designers and researchers with certain degrees of freedom to decide how they define design fiction's role in their work.

2.2 Design Fiction Methodologies

Design fiction employs a range of methodologies to explore and communicate speculative futures in a structured yet imaginative way. These methods help bridge the gap between current realities and fictional possibilities, often serving as tools for critical reflection, provocation, and discussion.

2.2.1 "What If" Questions and Scenarios

"What if" questions and scenarios are design fiction methodologies that explore potential futures and their implications. "What if" questions are hypothetical prompts that help start the exploration of different possible futures and initiate the imaginative process [2]. Meanwhile, "what if" scenarios are fictional narratives of the speculated futures, which complete the initial explorations done by the "what if" questions, to provide a relatively more concrete vision of the future [6]. The combination of "what if" questions and scenarios is a provocative way to spark discussion and exploration of possible futures. However, it is needed to note that the "what if" scenarios generated, while being fictional, should be ideally scientifically possible and based on social, cultural, ethical, and political implications of potential developments of our current world [2].

2.2.2 Design Fiction Matrix

The Design Fiction Matrix is a synthesis tool proposed by Vistisen [7] to help designers generate design fiction ideas and adopt design fiction in their designs. As design fiction is based on the linkage of facts and fiction and the speculation of the future would be built on what is happening in the present and what has happened in the past, Vistisen developed this matrix framework with a vertical "fact/fiction" axis and a horizontal "past/future" axis (see Figure 4). These two axes create four quadrants:

- Histories (fact/past quadrant): lessons we have learned from our past or present that the future scenarios are based on.
- Signals (fact/future quadrant): data or fact-based prediction and projection.
- Myths (fiction/past quadrant): myths or misunderstandings people might have that may impact how people think about the future.
- Scenarios (fiction/future quadrant): design fiction scenarios that speculate the future.

Vistisen believes that mapping these quadrants can help make the design fiction ideas more solid. As the matrix focuses on facts of the past and present, it pushes designers to rethink what assumptions they might have to prevent creating a biased or uninformed speculation of the future

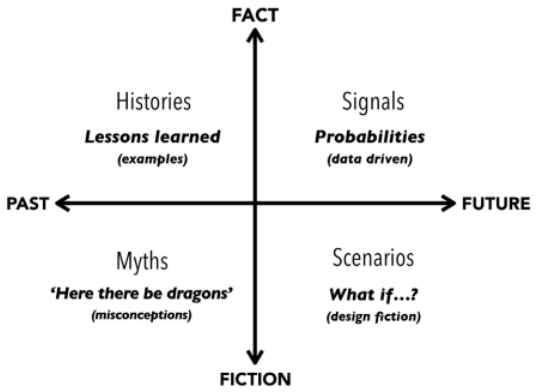


Figure 1: An illustration of the design fiction matrix with the fact/fiction and past/future axes creating the four quadrants.

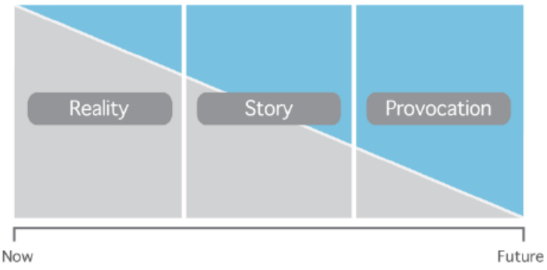


Figure 2: An illustration of this model with the three layers placed on the now/future axis with the proportion of fictional (blue) and factual (grey) contents each layer should consist of.

2.2.3 Three-Layer Model

Other than the three elements of design fiction mentioned previously, Lindley and Coulton [8] also proposed a three-layered model (see Figure 2) to contextualize the creation and communication of design fiction ideas, particularly facilitates the story creating and telling aspect of design fiction. This model consists of three layers:

- Reality layer: describes how the world today is known by referring to current phenomena, technologies, and knowledge.
- Story layer: narratives and plot devices that extract facts from the reality layer and extend them to a plausible future.
- Provocation layer: presents what will be designed in the future, that creates the discursive space for discussion.

These three layers can be placed on an axis from now to the future with the order of Reality, Story, and then the Provocation layer. When traveling from the extremes of now to the future, the proportion of fictional content gradually increases, while factual contents inversely decrease. However, regardless of which layer, there still consisted of both factual and future content. This model was developed based on a design fiction project, Lindley and Coul-

ton worked on, which uses film as the communication device.

2.3 Sound Design

Sound design requires a structured approach to shaping the story and conveying necessary information to the audience through sound. These aspects help create an immersive, contextually accurate, and engaging auditory experience.

2.3.1 Sound Layer

Frohlich [9] proposed and emphasized using multi-layered audiophotos to enrich the narrative of photos, enhancing memory recall and emotional connection. While William Whittington [10] identifies four fundamental principles that shape the use of sound in science fiction cinema. First, acoustic spectacle enhances the auditory experience through immersive and dynamic soundscapes to create a heightened auditory experience. Second, technological mediation utilizes sound as a representation of futuristic or alien technologies to alter perception. Third, psychological immersion enhances emotional engagement by influencing audience perception and intensifying narrative tension. Lastly, genre hybridization blends sound elements from various genres to create unique auditory experiences in sci-fi narratives. With these four principles, they indicate the crucial role of creating different layers to construct soundscapes in world-building and enhancing the sensory impact of the science fiction genre.

2.3.2 Categorization of Urban Environment Sounds

Tan et al. [11] explores a comprehensive classification method for common sounds in urban environments (see Figure 3). The study aims to establish a universal framework that aids soundscape researchers in the planning, preservation, and understanding of urban soundscapes. Through field recordings and participatory surveys, the research identifies various urban sounds and analyzes how people perceive and categorize them. Ultimately, the study proposes a classification system for urban sounds and examines the impact of different sounds on the urban soundscape experience.

This urban sound categorization study serves as a foundation for our future sound design by providing a structured framework for classifying and analyzing urban sounds. They systematically organize urban sounds and identify key urban sound elements, which reinforces our three-layer structure (human, mechanic, nature) and guides our sound composition. Since our future soundscapes are based on real-world developments, referencing an established classification system helps maintain contextual accuracy and realism, preventing the design from becoming purely speculative fiction.

2.3.3 Sound Localization

Hidaka et al.'s research [12] demonstrates that auditory information not only functions as an independent sensory channel but also significantly influences visual perception. They discovered that when a static visual stimulus is synchronized with alternating left-right sound sources, observers perceive the visual stimulus as moving laterally, which is known as Sound-Induced Visual Motion (SIVM). Changes in sound position affect our perception of a visual object's spatial location. This inspires us to position different sound sources at the four corners of our project and adjust volume based on movement distance, guiding users to explore the relationship between sound and space in the AR environment while influencing their perception and interaction with virtual objects.

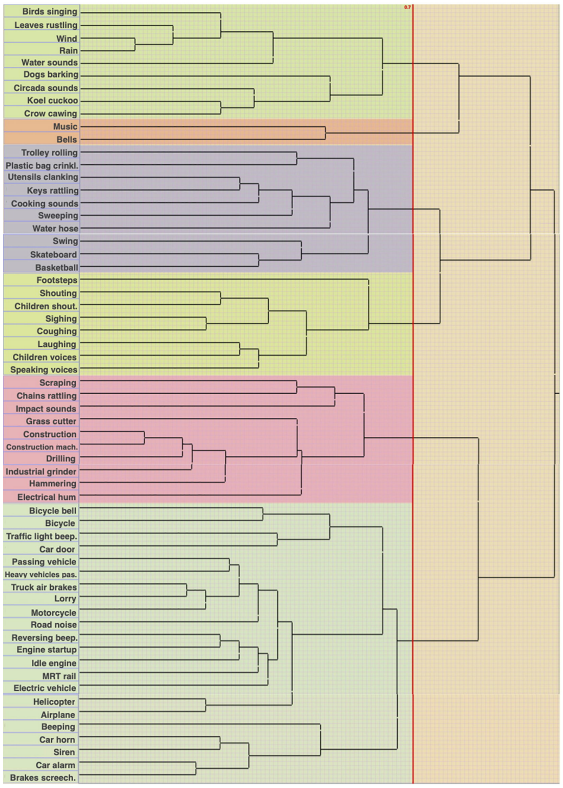


Figure 3: Hierarchical clustering of environmental sound categorization scheme of an urban city.

3. METHODS

Different methods were used in the design processes, development, and evaluation of this project. This includes the use of design fiction methodologies to create our vision of the possible future, how we created the soundscapes and the development of the AR interface. The methods were selected to support speculative exploration while grounding the work in real-world references.

3.1 Speculative Future with Design Fiction

In this project, we focus more on elements (1) and (3) presented by Lindley and Coulton (mentioned in Section 2.2.3) to design an interactive soundscape artifact that tells a story of our imagination of the future to inspire audiences to discuss their thoughts on the prospects of urban spaces. To do so, we used the design fiction matrix and "what if" questions and scenarios to imagine our speculation of future urban environments in Sweden, and then the three-layered model to decide how we want to communicate our vision of the future to the audiences.

3.1.1 *Imagination of the Future*

To create our imagination of the potential future, we first used the design fiction matrix mentioned in Section 2.2.2. We start on brainstorming relevant general phenomena that happened in the decade–such as global warming, the tense EU-Russia relationship, and the push for more sustainable technologies–to place in the Histories quadrant.

Then we research more specific events that we think are happening or might happen in the future. If the data or information shows that it is true, we place them in the Signals quadrant, if not, then we place them in the Myths quadrant.

Some of these events are relevant to the phenomena in the Histories quadrant:

- Global warming: decline in animal species and snow depth
- EU's right-wing political shift: stricter immigration policies, a reduction in the number of international students, and a decrease in the number of students admitted to higher education.
- Advancement of AI technology: the rise of driverless taxi.
- Push for more sustainable technologies: reduce in fossil fuel energy consumption, car companies' plans to produce only electric cars, more people prefer public transport

Meanwhile, some other events are less related to the phenomenon but are still relevant to urban space in Sweden, such as the birth rate in Sweden is declining while the population is increasing. We were able to find data or information to support all of the events, except for one event–a decrease in the number of students admitted to higher education, which turned out to be false after our research, and thus we placed this event in the Myth quadrant and all the other events in the Signals quadrant.

Based on the ideation and research we have done for the Histories, Myths, and Signals quadrants, we used the "what if" questions mentioned in Section 2.2.1 to create the future scenarios. Here are some of the questions to facilitate our imagination of the future:

- What if global warming continues to get more and more serious?
- What if the EU and Russia start a war?
- What if society becomes more conservative and anti-globalization?
- What if self-driving car technology becomes reliable?

3.1.2 *Storytelling of the Speculated Future*

To present our speculated future to the audience, we referenced the three-layered model mentioned in Section 2.2.3 to design our storytelling elements. Since this project does not aim to present a futuristic prototype like Lindley and Coulton's short film [8] but instead to tell the story of a possible future and spark discussion, we focus mainly on the Reality and Story layers. We decided to have two types

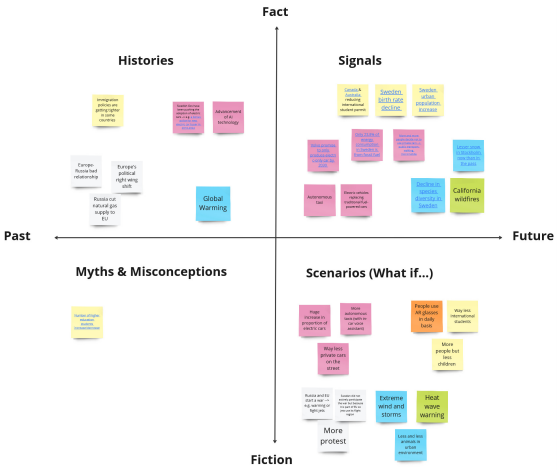


Figure 4: The design fiction matrix with the phenomena and events relevant to the future we speculate (*Link to Miro board*).

of soundscapes–present and future–with the present soundscape depicting the urban environment we know today (i.e. Reality layer) and the future soundscape illustrating how the future of urban spaces will be in 2040 (i.e. Story layer). By using these two types of soundscape to tell the story of the future through the Reality and Story layers, we hope to give the audience a glimpse of what the current urban space is like with the present soundscapes and have a contrast with the future soundscapes to understand what are the changes that might happen in the future.

Since, according to the three-layer model, all layers consist of both factual and fictional content, with the Reality layer having mostly factual content and the Story layer having a similar proportion of factual and fictional content. We decided to use mostly actual recordings of the KTH open-air square to compose the present soundscapes, with some additional sounds to show things we observed happening but not clearly shown in the recording or to emphasize aspects we would like to have a larger contrast with the future soundscapes. As for the future soundscapes, we still use the recordings as the basis, but we use more additional sounds and remove some parts of the recordings intentionally to create contrast from the present.

3.2 Soundscape Creation

To create the soundscapes, we used both actual recordings of the open-air square in KTH, sound clips, and synthesized sounds which are all edited in Audacity (see Figure 5).

For the recordings, the audio recorder we have used is the Zoom H4n Pro recorder, a 4-track recorder that allows at max 4 channels to be recorded at the same time. However, when immersive technologies, like AR, are paired with visual cues, stereo channel audio is enough to provide a sense of auditory source localization [13]. Therefore, we decided using the stereo mode of the recorder to record left and right channel tracks would be suitable. Since we hope the soundscapes can encapsulate what the urban space is like in general, we recorded four 5-minute clips of each corner

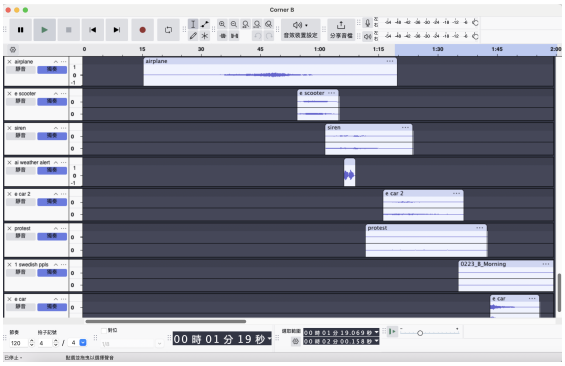


Figure 5: Interface of Audacity for the editing of the soundscapes).

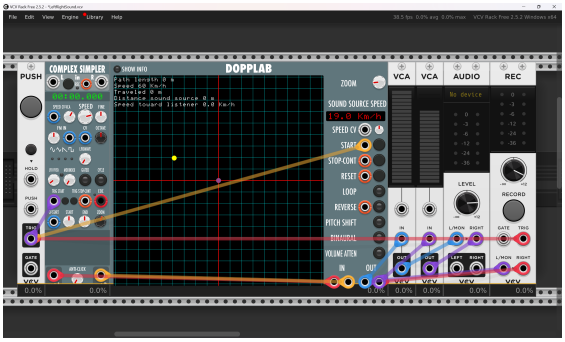


Figure 6: Interface of VCV Rack 2 for creating spacial audio through mono-channel sound clips.

of the square during the morning (10am), noon (12pm), afternoon (3pm), and evening (6pm) of different days, resulting in 20-minutes actual recordings for each corner of the square for creating the soundscapes.

Other than the recordings, we used sound clips from Freesound [14] and Sounds of Changes [15] with all sound clips under Creative Common Attribution or Attribution-NonCommercial licenses. As for the synthesized sound, VCV Rack 2 (see Figure 6) is used to create some synthesized sounds, like wind sounds, and used to create special audio effects for mono track audios.

3.3 Development

The AR experience was developed using Unity together with ARFoundation, this combination allows the development of across platforms AR applications. As we decided to use mobile devices, such as smartphones and tablets, for the AR experience, the application was deployed on Android devices, chosen for their accessibility for downloading external applications and support for ARCore. Instead of relying on external tracking and sensor systems, we used the device's built-in gyroscope sensors to detect orientation to enable rotational interaction (more details in Section **??**). The image marker (see Figure 7) is placed in the center of a controlled indoor environment for locating virtual anchors corresponding to photo markers to place the AR windows and sound source while preventing drifting. As for the spacial audio source, Unity has provided the Audio Source component that allows spacial audio by

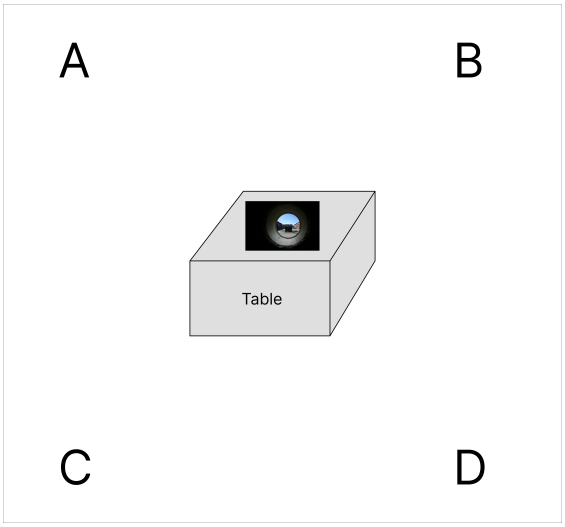


Figure 7: Illustration of where the image marker is placed in the center of the room.

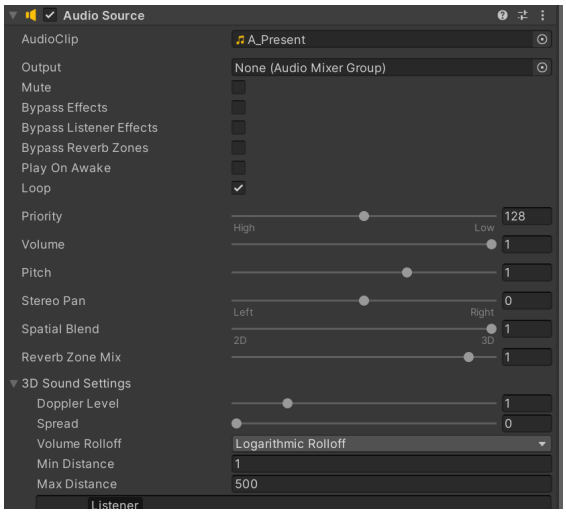


Figure 8: Audio Source component in Unity that supports spatial audio.

toggling the Spatial Blend value and the 3D Sounds Setting (see Figure 8).

3.4 Evaluation

Chung et al. [1] collected participants' imagination of the future through visual representations (i.e. sketches) in their experiment. The researchers asked participants to provide their visual interpretations of what they perceived. These sketches allowed the researchers to gain insight into the auditory imagination participants experienced while listening to soundscapes. These visual representations were considered to be imaginative and open to interpretation, facilitating reflection and re-imagination between researchers and participants. By using images to capture what participants heard, felt, and imagined while experiencing the future soundscape, the method not only helped the researchers understand participants' thoughts and responses, but also invited them to collaboratively complete the unfinished world-building, aligning with the open framework of design fic-

tion that encourages participants to imagine. Inspired by this approach, after participants experienced the soundscapes from the present to the future at the four corners, we invited them to describe or sketch their vision of the future soundscape, capturing what they heard and how they felt.

4. RESULTS

4.1 Soundscape Design

Based on the concept mentioned in 2.3.3, where different sound positions influence participants' spatial perception, we set up the scene in the AR environment, a square plaza with the fountain in front of the KTH Royal Institute of Technology D and E building. The soundscapes are positioned at the four corners of the plaza, labeled as A, B, C, and D (see Table 1). We used the audio recorded at each corner, as mentioned in 3.2. To convey the concept of design fiction, each corner has two soundscapes of now and the future, with the time set to 2025 and 2040, respectively. We listened carefully to the sounds in the recorded files, then, based on urban sound categorization and utilizing sound layers to tell the story, we formed three layers——human, mechanical, and nature (see Table 2). These three layers make up all the current and future soundscapes. The four corners emphasize different layers based on their positional characteristics (see Table 1). For example, A and B, being closer to the road, highlight the mechanical sounds more prominently. A, being closer to the park, emphasizes the Natural Layer, while B, near the sidewalk, emphasizes the Human Layer. Corner C, located next to a forest path within the campus, emphasizes the natural sounds, followed by the human layer, and then the mechanical sounds. D, situated at the campus entrance, is primarily a student pathway with a road through the campus, thus emphasizing the Human Layer, followed by the mechanical and natural layers.

In terms of sound, to highlight the difference between the present and the future, the current soundscapes are created by editing real recorded audio, and we have added necessary elements based on the contrast with the future (see Table 2). The future soundscapes, on the other hand, are created based on the design metrics outlined in section 3.1.1 to envision the future. In the human layer, there are sounds such as footsteps, speaking voices, and laughter, representing human activity. Based on the design metrics, trends such as urban population growth, low birth rates, and stricter immigration policies suggest that in the future, the human layer will feature more footsteps, fewer children's voices, and a lack of ethnic diversity, with only Swedish being heard in the soundscapes. Additionally, due to technological development and international tensions, as well as the potential for war, future audio includes sounds of humans operating AR interfaces and crowds participating in protests. In the mechanical layer, the sounds mainly come from various machines such as bicycles, oil cars, electric vehicles, and pedestrian signals. Notably, we believe that even after 15 years, pedestrian signals will remain unchanged, so this is one aspect of the current and future soundscapes that is not evolving. With the increasing population and the global shift toward electric vehicles, future soundscapes feature more electric vehicles, e-scooters, and self-driving taxis, along with military aircraft and war alert background sounds due to international tensions. Finally, in the nature layer, the present soundscapes include sounds like rain, wind, and bird calls. However, due to global warming leading to more extreme weather, the future soundscapes include sounds of heavy rain, storms, extreme weather warnings, and reduced bird calls resulting from the decline in biodiversity.

Corner Label	Sound Layers
A	++ Mechanic Layer + Natural Layer - Human Layer
B	++ Mechanic Layer + Human Layer - Natural Layer
C	++ Natural Layer + Human Layer - Mechanic Layer
D	++ Human Layer + Mechanic Layer - Natural Layer

Table 1: The four corners emphasize different layers based on their positional characteristics.

4.2 AR Installation

As mentioned in Section 3.3, image tracking of a photo with a tunnel leading to the fountain of the KTH square (see Figure 9) to place and anchor the window like AR objects–a frame with the photo of the corner (see Figure x)–on the four corners of the room to indicate the location of the sound sources. Windows is used as we hope to let the audiences know which corner of the KTH square they are listening to and allow them to imagine being in the KTH square. We originally planned to use a panoramic photo of the KTH square to create a higher immersion, however, it is harder and less safe to navigate the physical space of a room when the audiences are focused on the device without the physical environment displayed to them, and the image quality of the panoramic photo is relatively low which might instead reduce immersiveness.

When the windows are all placed in the four corners of the space, the audience can walk around the physical space to listen to the different soundscapes of each of the corners. We used the color of the window frames as an affordance to let audiences know how close and how loud they are hearing the sound of the corners. When the audience walks closer to the window at the corner, the frame becomes brighter in color (i.e. white) while the soundscapes of that corner get louder, and when they get further away the frame gets darker (i.e. black) and the volume gets lower (see Figure 10). If the audience is in between two corners, they will hear the soundscape of both of these corners spatially according to their location and orientation (i.e. they can hear the soundscape of different corners in the left and



Figure 9: The image of the fountain in the KTH square that acts as the image marker.



Figure 10: The color of the window frame changes to white when the audience is close to it and black when they are further away.

right channel) based on the idea of sound localization mentioned in Section 2.3.3 (see Figure 11).

As for the present and future soundscapes, we used the rotation of the device as the mechanism to transition between these two types of soundscapes. A clock with a text of present or future years placed in the middle is used as the interface to indicate the rotation. When the device is right-side-up, the present soundscapes are played with ”2025” text and the clock’s 12 o’clock mark painted in blue (see Figure 12). On the contrary, the future soundscapes will be played when the device is upside-down, with ”2040” text and the clock’s 6 o’clock mark painted in red (see Figure 12). When the device is rotated, the year text will increase or decrease, the clock mark painted in black will change, and the present and future soundscapes will get louder and softer gradually according to the rotation. This design follows two of the four key aesthetic interaction qualities proposed by Löwgren [16]–pliability (i.e. the feeling of a tightly coupled and responsive interaction) and fluency (i.e. the ease and grace of handling the changes between multiple demands and outcomes)–to ensure that the interaction is intuitive and aesthetically pleasing to the audiences.

When the audiences open the application for this experience, they will be greeted with a short introduction and a

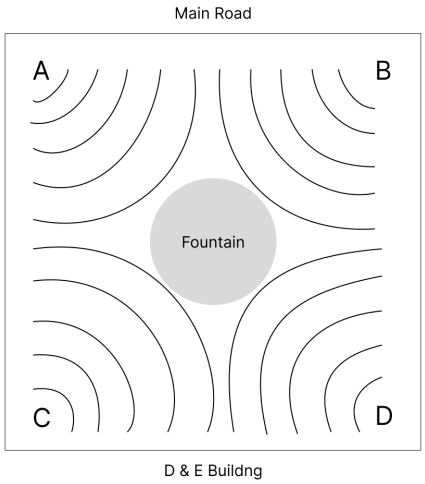


Figure 11: Illustration of how the sound source are placed in the four corners of the KTH square.



Figure 12: The clock interface when right-side-up at year 2025 and up-side-down at year 2040.

tutorial telling them that they will need to walk around the space to explore different soundscapes and rotate the device to switch between the present and future soundscapes (see Figure 13). After they finish the tutorial, there will be a prompt to ask the audience to scan the image to place the AR elements and start the experience.

4.3 User’s Speculation of the Future

We collected participants’ feedback with the evaluation method mentioned in Section 3.4. Participants’ feedback varied, with some expressing their thoughts through writing and others through drawings (see Figure 14). Regarding the sound, some mentioned “loud noise.” In terms of environmental imagination, responses included “too many people, too much technology, and protest sounds”, while others envisioned a “more natural future.” Emotionally, participants expressed feelings of “stress, pessimism, and discomfort”, but some also conveyed a sense of “hope.” In the drawings, many depicted airplanes, cars, and the use of AR interfaces. Additionally, some illustrations portrayed war, explosions, extreme weather, and the separation of loved

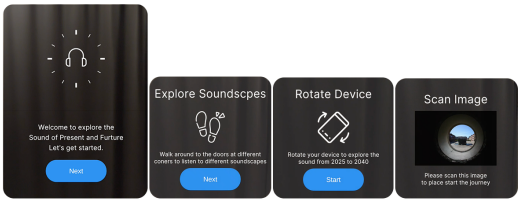


Figure 13: The greeting, tutorial, and scanning prompt interfaces are displayed when the AR application is first opened.



Figure 14: All the drawings and writings from the participants for their idea of urban future.

ones. On the other hand, there were also representations of a bright future and hopeful imagery.

5. DISCUSSION

5.1 Evaluation Findings

The evaluation of our experiment aimed to understand how participants perceived and interpreted the future soundscapes we designed. By analyzing both written and visual feedback, we gained insights into how sound influences future imagination, emotional responses, and perceptions of urban environments. In this project, we gained different aspects of the user feedback.

First, we can find that there are diverse interpretations of the future from the drawing and written feedback. Participants' responses revealed contrasting visions of the future, with some imagining a technologically advanced yet chaotic world (e.g. too many people, too much technology, protest sounds), while others envisioned a more natural and harmonious environment. Second, the soundscapes evoked different emotions. Many participants expressed negative emotions, such as stress, pessimism, and discomfort, suggesting that the projected future scenarios evoked anxiety. However, some participants also conveyed hope, indicating that the soundscapes allowed space for optimistic future possibilities. Thirdly, some of the scenes recur in visual and written responses. The dominant themes in participant sketches included airplanes, cars, AR interfaces, war, explosions, and extreme weather, aligning with the predicted future scenarios. Conversely, some participants illustrated a bright future, highlighting the subjective and open-ended nature of future imagination. Lastly, the soundscapes we designed effectively conveyed the future scenarios. The soundscapes triggered vivid mental imagery, supporting the idea that auditory elements can shape spatial and future perception. The mixed reactions indicate that sound-based storytelling can be a powerful yet subjective tool in design fiction, engaging participants in critical reflection on future possibilities.

The findings revealed a spectrum of reactions, ranging from anxiety and pessimism to hope and optimism, highlighting the diverse ways in which individuals engage with speculative auditory experiences. These results suggest that sound can effectively serve as a medium for speculative design, allowing participants to immerse themselves in imagined futures and express their interpretations through both text and visuals.

5.2 Limitation

The paper we feature for the urban design category is based in Singapore. While Singapore and Stockholm are both urban cities, soundscapes differ significantly due to variations in climate, transportation, and cultural activities. Therefore, we referenced sounds from the paper, such as footsteps, speaking, car horns, and bicycles, but things like cicada songs or koel cuckoo are sounds we don't have in Stockholm. We also add sounds that happened in Stockholm to make the scene more relatable, such as e-scooters and Swedish pedestrian signals.

Although the future scene is derived from the history and data of design metrics methodology, it is ultimately just an extension of our perspectives in this project. Not only is the sample size too small, but the information is also highly subjective. To further develop and make this project more complete, we could consider organizing a workshop to gather insights from more people as a foundation. Additionally, collecting more data and opinions would help create a future scenario that is more diverse and well-rounded.

There's also a gender imbalance in our speech data. In our recorded sound clips, most of the audible conversations were from male voices, which reflects KTH's gender ratio of 30% women and 70% men. This imbalance may have unintentionally influenced the human layer of the soundscape, making it less representative of a more diverse social environment in the present soundscapes.

The other challenge we faced was future soundscapes still grounded in reality. Since our future scenarios are set in a plausible near future (2040) rather than a distant sci-fi world, the sounds we designed remain relatively grounded in reality. While there are elements of technological advancement and societal shifts, the soundscape may not feel dramatically futuristic or vastly different from today's urban environment.

Layer	Sound Type	Present	Future
Human	Footstep	Based on the recording	More footsteps due to urban population increase
	Speaking Voice	More different languages (Chinese, German, Indian)	Only Swedish due to reduced international student permits
	Children Voice	Based on the recording	Less children voices due to decreasing birth rate in Sweden
	Laughing	Based on the recording	More laughing due to increased urban population
	AR Interface	N/A	Sound feedback from AR glasses as AI technology advances
	Protest Sound	N/A	Sounds of protests due to war and climate change activism
Mechanic	Bicycle	Based on recordings	A bit more, as more people use bicycles
	Traffic Light Beep	Important but hard to hear in recordings	No change, has been stable for 20 years
	Electric Car (Braking)	Based on recordings	More, as electric cars increase
	Gas Car (Braking)	Based on recordings	Almost none, replaced by electric cars and other transport
	Car Honk	Based on recordings	No significant change
	E-Scooters	Based on recordings	A bit more, as more people adopt e-scooters
	Aircraft	Some normal commercial airplane noise	More jets/war jets due to EU-Russia war
	Construction	Based on recordings	Balanced increase due to population growth and war influence
	Warning Signals	N/A	War warnings and extreme weather alerts
	Robotic/AI Voice	N/A	Voice assistance for autonomous taxis
	Car Brake Screech	Based on recordings	Less, as electric cars produce less screeching noise
Nature	Rain	Based on the recording	Heavier rain due to climate change
	Wind	Based on the recording	More extreme winds and storms
	Birds Singing	More birds chirping	Less birds due to species decline in urban areas
	Leaves Rustling	Based on the recording	Trees will continue to grow and wither, maintaining rustling sounds

Table 2: This table categorizes environmental sounds into three layers ——Human, Mechanic, and Nature, and compares the possible changes in sounds between the present and the future.

6. CONCLUSION

This project set out to explore how sound can serve as a medium for reflection on the speculated future of the urban space. By blending design fiction methodologies with AR and spatial audio, we created an interactive experience that allows audiences to engage with both the present and imagined future soundscapes of a familiar urban location. Through layering different categories of sonic elements and intuitive AR interactions, we encouraged participants to consider how urban environments may evolve in response to technological, environmental, and sociopolitical changes.

Our findings indicate that sound is not only a storytelling tool but also a prompt for reflection and imagination. Participants' emotional and cognitive responses—ranging from anxiety and discomfort to hope and curiosity—demonstrate that immersive soundscapes can provoke meaningful interpretations of future possibilities. Furthermore, the use of sound design with design fiction grounded in real-world phenomena helped maintain a balance between imagination and plausibility, reinforcing the value of sound in shaping perceptions of space and time.

Ultimately, this project contributes to conversations around urban futures, interactive storytelling, and the potential of sound as a medium for design fiction. It highlights how emerging technologies can invite more participatory and multi-sensory engagement with questions of what our cities might become, and how we might listen to the world differently—both now and in the years to come.

7. REFERENCES

[1] W.-M. D. Chung, W.-C. Tsai, R.-H. Liang, M. Liu, B. Kong, Y. Huang, and F.-C. C. and, "Soundscape fiction: designing and situating sonic design fiction to stimulate the imagination," *Digital Creativity*, vol. 0, no. 0, pp. 1–23, 2024. [Online]. Available: https://doi.org/10.1080/14626268.2024.2428920

[2] S. Lundmark, M. Jonsson, M. Myhre, A. Hjuberger, and R. Rodela, "Listening to the future: Using participatory sound fiction to engage young people in urban design," in *Proceedings of the 26th International Academic Mindtrek Conference*, ser. Mindtrek '23. New York, NY, USA: Association for Computing Machinery, 2023, p. 223–232. [Online]. Available: https://doi.org/10.1145/3616961.3616981

[3] B. Sterling, "Sci-fi writer bruce sterling explains the intriguing new concept of design fiction (interview by torie bosch)," 2012, accessed: 2025-03-07. [Online]. Available: http://www.slate.com/blogs/future_tense/2012/03/02/bruce_sterling_on_design_fictions_.html?via=gdpr-consent

[4] J. Bleecker, *Design Fiction*. John Wiley Sons, Ltd, 2022, ch. 24, pp. 561–578. [Online]. Available: https://onlinelibrary.wiley.com/doi/abs/10.1002/9781119815075.ch47

[5] J. Lindley and P. Coulton, "Back to the future: 10 years of design fiction," ser. British HCI '15. New York, NY, USA: Association for Computing Machinery, 2015, p. 210–211. [Online]. Available: https://doi.org/10.1145/2783446.2783592

[6] K. Baumann, B. Stokes, F. Bar, and B. Caldwell, "Infrastructures of the imagination: Community design for speculative urban technologies," in *Proceedings of the 8th International Conference on Communities and Technologies*, ser. CT '17. New York, NY, USA: Association for Computing Machinery, 2017, p. 266–269. [Online]. Available: https://doi.org/10.1145/3083671.3083700

[7] P. Vistisen, "The design fiction matrix: A synthesis tool for grounding fiction scenarios in facts from reality," vol. 2021. Nordes Digital Archive, 2021, pp. 125–129, nORDES 2021 ; Conference date: 15-08-2021 Through 18-08-2021. [Online]. Available: http://conference2021nordes.org

[8] J. Lindley, "Modelling design fiction: What's the story?" 06 2014.

[9] D. M. Frohlich, *Audiophotography : Bringing Photos to Life with Sounds*, 1st ed., ser. Computer Supported Cooperative Work, 3. Dordrecht: Springer Netherlands, 2004.

[10] W. WHITTINGTON, *Sound Design and Science Fiction*. University of Texas Press, 2007. [Online]. Available: http://www.jstor.org/stable/10.7560/714304

[11] J. K. A. Tan, Y. Hasegawa, and S.-K. Lau, "A comprehensive environmental sound categorization scheme of an urban city," *Applied Acoustics*, vol. 199, p. 109018, 2022. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0003682X22003929

[12] S. Hidaka, Y. Manaka, W. Teramoto, Y. Sugita, R. Miyauchi, J. Gyoba, Y. Suzuki, and Y. Iwaya, "Alternation of sound location induces visual motion perception of a static object," *PloS one*, vol. 4, no. 12, pp. e8188–e8188, 2009.

[13] C. C. Berger, M. Gonzalez-Franco, A. Tajadura-Jiménez, D. Florencio, and Z. Zhang, "Generic hrtfs may be good enough in virtual reality. improving source localization through cross-modal plasticity," *Frontiers in Neuroscience*, vol. Volume 12 - 2018, 2018. [Online]. Available: https://www.frontiersin.org/journals/neuroscience/articles/10.3389/fnins.2018.00021

[14] "Freesound." [Online]. Available: https://freesound.org/

[15] "Sounds of changes." [Online]. Available: https://www.soundsofchanges.eu/

[16] J. L. and, "Toward an articulation of interaction esthetics," *New Review of Hypermedia and Multimedia*, vol. 15, no. 2, pp. 129–146, 2009. [Online]. Available: https://doi.org/10.1080/13614560903117822